

Query-Prompt Evaluation — Production vs. Updated (Test) Prompts

A/B evaluation of two query-prompt versions under an identical system prompt, to test whether the updated prompts produce **less verbose** responses without losing factual quality or citations. `cp-ai-rag-service` — harness module `ai-document-system-prompt-harness-eval`.

1. What was tested

Two versions of the same 10 query prompts, matched by `queryId` :

- **prod** — mirrors the production `queryPrompt` for each query.
- **test** — rewritten with explicit output structure and, in several cases, hard word limits, aiming for shorter answers.

Both versions were run through the full production pipeline (embed → AI Search → refinement → LLM → citation post-processing) under **one fixed system prompt** (`v4-strict-citation-grouping-compact`), so any difference is attributable to the query prompt alone.

Run configuration

Setting	Value
System prompt	<code>v4-strict-citation-grouping-compact</code> (only)
Query versions	<code>prod</code> , <code>test</code> (same 10 queries, joined by <code>queryId</code>)
Documents	2 (<code>4fa52386...</code> , <code>22e543b6...</code>)
Models	<code>gpt-4o-response-generation</code> , <code>gpt-5.1</code>
Repetitions	2
Generation calls	10 × 2 versions × 2 docs × 2 models × 2 reps = 160
Retrieval	kNN 80 → pool 60 → containment-dedup + MMR → 30 chunks to the LLM
Citation guard	<code>off</code> (measurement mode — degraded answers are measured, not thrown)
Quality judge	<code>gpt-5.1</code> , 80 pairwise <code>prod↔test</code> judgements
Reasoning effort	<code>none</code> (<code>gpt-5.1</code>) • max tokens 7,000

Mechanics: 160/160 generation calls and 80/80 judge calls succeeded, zero errors.

Each answer's metrics are averaged over its 2 documents × 2 repetitions (n = 4 per cell). Two citation metrics are distinguished throughout:

- **cited pages** — pages listed in the model's `FACT_MAP_JSON` block (evidence the model *identified*).
- **rendered citations** — inline source markers actually *shown to the reader* after post-processing.

These usually agree, but they diverge in the `gpt-4o` citation regression (§4): the model identifies the evidence (cited pages present) but omits the inline markers, so the citations are stripped and never rendered. **uncited** counts substantive answers (≥ 50 prose words) with **zero rendered** citations. The judge scores each answer against **its own** instruction (so a test prompt that caps length is not penalised for omitting detail it was told to omit).

2. Headline summary

Verbosity — the test prompts achieve their goal

Model	prod words/answer	test words/answer	Change
gpt-5.1	~952	~586	−38%
gpt-4o	~274	~215	−22%

Coverage held up far better than length fell: on gpt-5.1, cited pages dropped only ~19% against the 38% word cut — denser answers, not thinner evidence. (On gpt-4o the model still *identified* as much evidence, but for the test version much of it was not rendered to the reader — the citation regression in §4.)

Factual quality — the shorter answers are equal-or-richer

Judge verdicts, prod (A) vs test (B), 40 pairs per model:

Model	EQUIVALENT	prod-richer	test-richer	DIVERGENT	Structure prod→test
gpt-4o	14	10	13	3	4.6 → 4.8
gpt-5.1	16	8	13	3	4.5 → 4.8

The test version is judged **equal-or-richer more often than it is poorer**, on both models, while improving adherence to the requested format.

Citations — one model-specific regression

- **gpt-5.1: perfect on both versions** — 0 stripped markers, 0 uncited answers across all 80 answers.
- **gpt-4o + test: 15 of 40 substantive answers rendered with zero citations** — concentrated in exactly four queries (§4). Root cause is a missing `[citation]` cue in those test prompts' output templates, not a pipeline fault.

Bottom line

Two separate questions, two answers:

1. **Prompt rewrite (same model):** the test prompts are equal-or-better than prod on both models, gated only by the gpt-4o citation fix — details in §3, verdict in §6.
2. **Migration:** the target configuration (**test prompts on gpt-5.1**) beats current production (**prod prompts on gpt-4o**) on **every** query, 30 richer / 10 equal / 0 worse — §5.

3. Per-query analysis

Legend: **words** = mean prose words (prod → test); **cited pages** prod → test; **uncited** = of 4 answers, how many were substantive but had no citation; **judge** = test-vs-prod verdict over the 8 pairs for that query (both models, both docs, both reps).

3.1 Substantive analytical queries

Summary of the facts of the offences —  test wins

Model	words (prod→test)	cited pages	uncited (test)	citations
gpt-4o	706 → 526 (−25%)	13.5 → 13.5	0/4	intact both versions
gpt-5.1	2,415 → 1,199 (−50%)	24 → 16	0/4	intact both versions

Judge (8 pairs): **test-richer 6, prod-richer 2**, structure 4.5 → **5.0**. The single biggest verbosity win (gpt-5.1 halved) and the judge still rates the shorter test answers richer. The test prompt keeps an explicit "includes citations" instruction, so citations survive on both models. **Clear win.**

Chronology of the case — ⚠️ **more concise but loses procedural detail; gpt-4o citation regression**

Model	words (prod→test)	cited pages	uncited (test)	citations
gpt-4o	485 → 266 (−45%)	6.0 → 6.2	4/4	lost on test (gpt-4o)
gpt-5.1	2,126 → 953 (−55%)	25.5 → 12.2	0/4	intact

Judge (8 pairs): **prod-richer 7, divergent 1**. This is the one query where prod is judged *richer* — the test prompt drops investigative/procedural chronology (arrests, searches, interview content) that prod retained. Combined with the gpt-4o citation loss, **the test version needs revision here** before adoption: restore a citation cue and reconsider the procedural-events scope.

Applications on the case — ⚠️ **good compression; gpt-4o citation regression**

Model	words (prod→test)	cited pages	uncited (test)	citations
gpt-4o	159 → 74 (−53%)	2.8 → 1.5	3/4	lost on test (gpt-4o)
gpt-5.1	337 → 148 (−56%)	3.8 → 3.2	0/4	intact

Judge (8 pairs): prod-richer 4, equivalent 2, test-richer 1, divergent 1. Strong length reduction with mostly-equivalent content on gpt-5.1; the gpt-4o citation loss is the blocker.

Summary of prosecution evidence — ✅ **test wins (gpt-5.1); gpt-4o citation regression**

Model	words (prod→test)	cited pages	uncited (test)	citations
gpt-4o	579 → 343 (−41%)	16.2 → 11.8	4/4	lost on test (gpt-4o)
gpt-5.1	2,095 → 795 (−62%)	22 → 22.2	0/4	intact

Judge (8 pairs): **test-richer 6, prod-richer 2**, structure 4.9 → **5.0**. On gpt-5.1 this is an excellent result — 62% shorter, *same* page coverage, judged richer. The gpt-4o citation regression is the only issue.

Summary of each witnesses evidence — ✅✅ **best overall result**

Model	words (prod→test)	cited pages	uncited (test)	citations
gpt-4o	462 → 600 (+30%)	12.8 → 26.5	0/4	intact (fixes a prod pathology)
gpt-5.1	1,858 → 2,027 (+9%)	17.5 → 25.8	0/4	intact

Judge (8 pairs): **test-richer 7, divergent 1**, structure 4.1 → **5.0**. The exception to the verbosity trend — here the test prompt is slightly *longer* but markedly richer and better structured, roughly doubling gpt-4o page coverage. Critically, the test prompt **fixes the prod gpt-4o "witness-bundle" pathology** (prod averaged 250 stripped markers / had an uncited answer; test has 0). The test prompt retained "citation links" wording. **Strongest single win.**

Summary of the defendants previous convictions — test richer (gpt-5.1); gpt-4o citation regression

Model	words (prod→test)	cited pages	uncited (test)	citations
gpt-4o	223 → 242	7.8 → 9.5	4/4	lost on test (gpt-4o)
gpt-5.1	525 → 570	15.8 → 7.8	0/4	intact

Judge (8 pairs): **test-richer 5**, equivalent 2, divergent 1. The restructured test prompt (numbered sections 1–8) is judged richer and far better organised. gpt-4o citation regression again.

3.2 Fixed-list extraction queries (controls)

The prompts for **dwelling burglary, drug trafficking/supply, offensive weapon/blade, and alcohol/drug driving** were **identical** between prod and test (only the six analytical prompts changed). They act as a control set — any difference here is run-to-run noise.

Query	judge (8 pairs)	cos	Notes
Previous dwelling burglary	EQ 5, prod 2, test 1	0.97	Data-driven list; both versions equivalent
Previous drug trafficking/supply	EQ 8	1.00	No relevant convictions → identical short refusals
Previous offensive weapon/blade	EQ 5, divergent 2	0.99	Equivalent; minor run variance
Previous alcohol/drug driving	EQ 8	1.00	No relevant convictions → identical short refusals

The near-perfect equivalence and ~1.0 cosine on the identical prompts confirm the harness and the retrieval path are stable, so the differences seen on the analytical queries are real prompt effects, not measurement noise.

4. The gpt-4o citation regression — root cause

15 of 40 substantive gpt-4o test answers rendered **zero** citations. They map **exactly** to the four analytical queries whose updated `OUTPUT FORMAT` template contains **no [citation] placeholder**:

Query	test prompt has [citation] cue?	gpt-4o test uncited	gpt-5.1 test uncited
Chronology of the case	✗	4/4	0/4
Applications on the case	✗	3/4	0/4
Summary of prosecution evidence	✗	4/4	0/4
Summary of previous convictions	✗	4/4	0/4
Summary of the facts	✓ ("includes citations")	0/4	0/4
Summary of witnesses evidence	✓ ("citation links")	0/4	0/4

gpt-4o follows the query-level output template literally and lets it override the system prompt's citation contract; gpt-5.1 does not — it honours the system prompt regardless. The two test prompts that kept a citation cue stayed fully cited on both models, which confirms the mechanism.

Important for production: because these omissions are *instruction-induced and deterministic*, the citation guard's retry would not help — a redelivered attempt follows the same template and omits citations again. In

DELIVER mode they would ship uncited (flagged); in REJECT mode those four queries would consistently fail on gpt-4o. **The fix belongs in the query prompts, not the pipeline.**

5. Migration view — production today vs proposed target

This is the actual deployment decision: **current production** (`prod` query prompt on **gpt-4o**) against the **proposed target** (`test` query prompt on **gpt-5.1**). Unlike §3 (which varies only the prompt), this varies both prompt and model together. The judge scored 40 pairs over the same generated answers (10 queries × 2 documents × 2 reps), each answer against its own instruction.

The target wins outright

Judge verdict (target vs production)	Pairs
target (test + gpt-5.1) richer	30 / 40
Equivalent	10 / 40
production (prod + gpt-4o) richer	0
Divergent	0

Every substantive query was judged target-richer in all 4 of its pairs; the only ties were the two genuine no-conviction queries. Structure adherence rose from ~3.5–4.8 to **5.0** almost everywhere. **Note the important nuance:** the Chronology result that looks like a *prod* win in §3 (a same-model finding) **reverses here** — gpt-5.1's stronger extraction more than compensates for the test prompt's trimmed scope, so the target beats production on Chronology too.

The trade

	production (prod + gpt-4o)	target (test + gpt-5.1)	Change
Words / answer	~274	~586	+114%
Cited pages (total, 40 answers)	264	375	+42%
Rendered citations (total)	~200	~770	~3.9×
Uncited substantive answers	2	0	eliminated
Stripped (lost) markers	1,202	0	eliminated

The migration buys **~3.9×** the user-visible citations, better grounding, and the removal of gpt-4o's **citation pathologies**, at the cost of **~2×** longer answers. The citation-cue problem that blocks the test prompts on gpt-4o (§4) **does not arise here** — gpt-5.1 honours the system prompt's citation contract regardless of the query template.

6. Verdict and recommendations

The two questions, side by side

	Q1 — Prompt rewrite (same model)	Q2 — Migration (test + gpt-5.1 vs prod + gpt-4o)
Evidence	§3	§5
Winner	Test on gpt-5.1; mixed on gpt-4o (citation bug)	Target, on every query (30 richer / 10 equal / 0 worse)

The per-query "same-model winner" table below answers **Q1** — it is diagnostic (it tells you which test prompts still need work). It does **not** override the migration answer: for **Q2**, the target configuration wins every query (§5).

Same-model winner (Q1)

Verbosity Δ on gpt-5.1. "Judge" is the net test-vs-prod verdict over the 8 same-model pairs (both models, docs, reps). * = test wins once its [citation] cue is restored (§4).

#	Query	Verbosity Δ (gpt-5.1)	Judge (test vs prod, same model)	gpt-4o citations	Same-model winner
1	Summary of the facts	−50%	test-richer 6 / prod 2	✓ intact	● Test
2	Chronology of the case	−55%	prod-richer 7 / div 1	✗ lost	● Prod — test over-trimmed + needs citation fix
3	Applications on the case	−56%	prod 4 / EQ 2 / test 1	✗ lost	● Prod (narrow) — test after citation fix
4	Summary of prosecution evidence	−62%	test-richer 6 / prod 2	✗ lost	● Test*
5	Summary of each witnesses evidence	+9%	test-richer 7 / div 1	✓ intact (fixes prod bug)	● Test
6	Summary of previous convictions	+9%	test-richer 5 / EQ 2	✗ lost	● Test*
7–10	The four fixed-list queries (controls)	~0 (identical prompts)	mostly EQ	✓ intact	● Tie

Same-model tally: Test wins 2 outright (Facts, Witnesses), 2 after the citation fix (Prosecution evidence, Convictions); Prod retains 2 (Chronology on content, Applications narrowly); 4 controls tie.

Per-query changes to make the test prompts production-ready

All six substantive queries are target-richer 4/4 in the migration (§5), so these are **hardening** changes — the priority is making the test prompts citation-safe on gpt-4o and restoring the one scope that was genuinely over-trimmed. ★ = required for gpt-4o safety.

Query	Change to consider
Summary of the facts	Strong as-is (keeps a citation cue). Optional: add a word cap (≤ 600) for consistency.
Chronology of the case	★ Add the [citation] cue; restore procedural/investigative events (arrests, searches, interviews, bail) — the one scope the rewrite over-trimmed. Keep the concise bullet format.
Applications on the case	★ Add [citation] to the bullet template. Otherwise excellent compression (−56%); no other change.
Summary of prosecution evidence	★ Add [citation] to the "Key Evidence" / "Defendant Account" lines. Keep the 550-word cap and per-charge structure.
Summary of witnesses evidence	Keeps citation links + 200/500-word caps. Optional: the 500-word total cap may bind on very multi-witness cases — consider per-witness only.
Summary of previous convictions	★ Add [citation] to the numbered sections (the 8-section rewrite dropped prod's trailing cue). Keep the structure — it scored 5.0.
The four fixed-list queries	No change — prompts identical to prod and already carry [citation].

Recommendations

1. **Adopt the target configuration (test prompt + gpt-5.1).** Judged richer-or-equal in 40/40 pairs, never worse, $\sim 3.9\times$ the rendered citations, no gpt-4o citation defects — at the cost of $\sim 2\times$ length.
2. **★ Apply the citation-cue fix** to the four test prompts (Chronology, Applications, Prosecution evidence, Previous convictions). Required only for gpt-4o, but cheap and makes every prompt model-agnostic — do it regardless, so gpt-4o remains a safe fallback.
3. **Restore the Chronology procedural scope** — the sole content the rewrite genuinely lost.
4. **Keep the length caps and heading structure** — they drive the verbosity reduction and the 5.0 structure scores at no quality cost.
5. **Re-verify cheaply** — a gpt-4o-only, test-version-only pass (40 calls) after the edits, to confirm the 15 uncited answers go to zero.

Generated from this round's harness run: 160 gpt-4o + gpt-5.1 generation calls; 80 gpt-5.1 judge pairs for the within-model prod $\leftrightarrow$ test comparison (§2–§4, §6) plus 40 gpt-5.1 judge pairs for the production-vs-target migration comparison (§5), judged over the same generated answers; 2 documents, 2 repetitions, v4 system prompt. Metrics are averages over documents $\times$ repetitions; judge verdicts are counts over the pairs per query.